

Research Paper

A unified contact model incorporating surface abrasion for DEM simulation of granular soil behavior across small to large strains

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ABSTRACT

The discrete element method (DEM) is a valuable tool for simulating the mechanical behavior of granular soils at both large and small-to-medium strains. However, the use of different contact models for varying strain levels has led to inconsistencies in previous simulations. This study highlights that while factors such as particle shape and fragmentation are critical, the omission of surface degradation mechanisms in current models may significantly contribute to their unsatisfactory performance in conventional undrained simulations. To address this, a unified contact model is proposed to account for the influence of surface degradation (specifically abrasion) on the soil behaviors across small to large strains. The proposed model is formulated such that the normal force–displacement law transitions naturally from Hertzian contact mechanics at small strains to a linear-bounded constitutive relation at large strains, capturing the accumulation of surface degradation as a primary driver of the soil skeleton's increased compressibility. Simulations confirm that the proposed model effectively captures the macroscopic mechanical behavior of granular soils at the considered strain levels, showing good agreement with experimental data. In addition, the comparative analyses of macroscopic and microscopic properties are revealing that small strain modulus G_0 and macro state parameter ψ_{e0} can be well correlated with mechanical coordination number MCN and micro state parameter ψ_{MCN0} , respectively.

1. Introduction

The stress–strain relationship of granular soils is highly nonlinear. Studies show that soil behavior is nearly fully recoverable at small strains, typically defined as shear strains below 10^{-6} for sands. In contrast, the large-strain regime refers to shear strains exceeding 1×10^{-3} , a range that can be reliably measured in conventional soil testing (e.g., triaxial or oedometer tests) without special instrumentation (Atkinson, 2000; Burland, 1989; Castellón and Ledesma, 2022; Clayton, 2011; Benz, 2007). The discrete element method (DEM) is a powerful tool for modeling such particulate materials across these varying strain magnitudes (Cundall, 1988; Cundall et al., 1989; Cundall and Strack, 1979; Ng and Petrakis, 1996; Thornton, 2000). However, a significant challenge in DEM arises from a dichotomy in contact model selection. For large-strain undrained simulations using the constant volume method, the simple linear contact model is usually adopted for successfully reproducing monotonic laboratory responses (Dubujet and Dedecker, 1998).

In contrast, for small-strain behavior, the Hertz contact model is essential for capturing the nonlinear evolution of the small-strain modulus G_0 with pressure (Yimsiri and Soga, 2000). When using a realistic particle shear stiffness for quartz sand (e.g., 29 GPa), the Hertz model accurately replicates results from bender element and resonant column tests (Gu et al., 2020a; Gu and Yang, 2013). However, the Hertz model with the same particle stiffness yields unrealistically high deviatoric stress and pore water pressure in constant-volume undrained simulations (Hanley et al., 2013), creating a critical inconsistency in modeling practices for granular materials. The inapplicability of the Hertz model with this realistic particle stiffness in undrained simulations indicates that effective factors have not yet been adequately considered. While particle geometry and fragmentation are critical drivers of the mechanical behavior, this study postulates that surface degradation mechanisms are essential to bridge the gap between grain-scale stiffness and bulk skeleton compressibility.

To resolve this issue, previous studies have explored modifications to

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Table 1

The results obtained from isotropic compression tests.

Confining pressure (kPa)	Void ratio	Bulk modulus of soil skeleton K_{sk} (GPa) / Pore water pressure coefficient B^*		
		rr-linear ^{**} (Gu et al., 2020b)	rr-Hertz ($G_g = 1.46$ GPa)	rr-Hertz ($G_g = 29$ GPa)
100	0.731	0.035 / 0.990	0.021 / 0.994	0.143 / 0.962
100	0.661	0.050 / 0.986	0.025 / 0.993	0.182 / 0.950
1000	0.733	0.038 / 0.990	0.043 / 0.988	0.213 / 0.945
1000	0.658	0.052 / 0.985	0.053 / 0.985	0.283 / 0.925

* The bulk modulus of typical saturated sand ranges from 0.011 to 0.056 GPa, with a B -value above 0.98 (Mitchell and Soga, 2005; Head, 1992);

** $k_0 = 8 \times 10^8 \text{ N/m}^2$ and $\alpha = 0.15$.

the Hertzian contact framework. Approaches incorporating surface roughness, such as the semi-empirical Greenwood and Williamson model, have shown that the consideration of grain-scale features can increase large-strain shear strength (Greenwood et al., 1984; Greenwood and Williamson, 1966; Zhao et al., 2018). Other enhancements include more tailored pressure distribution at particle level and plastic deformation due to particle asperity (Li et al., 2000; Nie et al., 2025; Otsubo et al., 2017; Sandeep and Senetakis, 2018). Although these models acknowledge that factors include surface roughness influence particle stiffness at small strains, they have not yet produced a single coherent model suitable for all interested strain levels.

While numerous insightful models have been proposed, developing a coherent framework for simulating granular soils across the complete strain spectrum remains an important goal. Previous studies have provided valuable avenues for reconciling the discrepancies between different contact models. For instance, Keishing and Hanley (2020) investigated the effects of particle fragmentation, surface roughness evolution (under normal contact), and water–air mixture compressibility on undrained Hertz model simulations. They concluded that these factors provided only limited improvements to the simulations. Another effective approach is the adjustment of the particle shear stiffness (e.g., $G_g = 1.46$ GPa) of the Hertz model, which successfully yields reasonable deviatoric stress and pore water pressure for monotonic undrained shearing (Phan et al., 2021; Wu et al., 2022; Yang and Huang, 2024; Zhang and Wang, 2015). This method, while effective for large-strain analysis, presents challenges for maintaining physical realism at the small-strain scale, where the shear stiffness of quartz sand particles is known to be much higher. Other promising lines of inquiry include incorporating the effects of pore fluid compressibility through poroelasticity and developing contact models that account for stiffness degradation (Kuhn and Daouadji, 2020; Wu et al., 2022). While these two approaches offer improvements under specific conditions, their simulation results remain unsatisfactory from small to large strain. Despite these significant advances, challenges still persist in creating a physically sound model that unifies soil behavior across small, intermediate, and large strains. Further investigation is warranted to develop a more comprehensive contact model capable of emulating the mechanical response of granular soils for the entire strain range.

To address the aforementioned challenges, this study presents a systematic investigation into the mechanical behavior of granular soils. First, we analyze the possible underlying mechanisms responsible for the discrepancies that arise when using the Hertzian model in constant-volume undrained tests. Building on the analysis, we propose and implement a unified contact model that incorporates the particle abrasion effect to capture the soil behaviors across the full strain spectrum. To ensure the performance of the proposed model, it is calibrated against experimental data to realize satisfactory results. Finally, the

work provides a comprehensive analysis of the relationship between macroscopic mechanical behavior and the evolution of microstructure from small to large strains.

2. Soil skeleton compressibility using classical established contact models

This study uses Itasca's Particle Flow Code (PFC^{3D}) to carry out DEM simulations (Itasca, 2018). To investigate the unsatisfactory performance of the Hertz model in monotonic undrained tests, we first performed a series of isotropic compression simulations using different classical contact models. This section details the numerical setup and presents an analysis of the soil skeleton's bulk compression modulus, which is a key factor in simulating undrained behavior.

The specific models used are the standard linear, rolling-resistance linear (rr-linear), and Hertz models, whose formulations are detailed in the Itasca Software Documentation (Itasca, 2018). A rolling-resistance Hertz (rr-Hertz) model is also used to account for particle shape effects in a computationally efficient manner, where the rotational stiffness is related to the tangential stiffness in the same manner as the rr-linear model (Iwashita and Oda, 1998).

The numerical specimen was created to emulate Toyoura sand. The particle size distribution ranges from 0.1 mm to 0.43 mm, with a mean particle size d_{50} of 0.16 mm and a coefficient of uniformity C_u of 1.46 (Fukushima and Tatsuoka, 1984). A total of 44,488 spherical particles were generated within six rigid walls using the particle expansion method, and the sample was pre-compressed to a confining pressure of 10 kPa. Different initial void ratios were achieved by varying the friction μ and rolling resistance μ_r coefficients during pre-compaction. Following this stage, these coefficients were set to their final values for all subsequent consolidation and shearing tests. After consolidation, the specimen was subjected to a series of subsequent tests (i.e. isotropic compression, probe, drained, and undrained tests). A more detailed description of the specimen generation and input parameters can be found in Gu et al. (2020b).

A series of strain-controlled isotropic compression tests was performed on consolidated specimens with similar void ratios to analyze the discrepancies between different contact models. For the rr-Hertz model, two different particle shear moduli were used: a physically realistic value of $G_g = 29$ GPa and a much lower value of $G_g = 1.46$ GPa. In all models, the particle Poisson's ratio ν_g was 0.15, while the friction μ and rolling resistance μ_r coefficients were 0.45 and 0.5, respectively. The tests were terminated when the volumetric strain reached 1×10^{-3} .

Given an isotropic granular assembly, the bulk compression modulus of the soil skeleton can be obtained as

$$K_{sk} = \frac{\Delta p'}{\Delta \varepsilon_v} \quad (1)$$

where p' is the mean effective stress and ε_v is the volumetric strain.

In the triaxial test, the Skempton pore water pressure coefficient B can be obtained as (Bishop, 1973; Skempton, 1954)

$$B = \frac{1}{1 + n \frac{C_w - C_g}{C_{sk} - C_g}} \approx \frac{1}{1 + n \frac{C_w}{C_{sk}}} = \frac{1}{1 + n \frac{K_w}{K_{sk}}} \quad (2)$$

where C_w denotes the compressibility of water, C_g denotes the compressibility of the particle, C_{sk} denotes the compressibility of the soil skeleton, K_w denotes the bulk modulus of water (i.e. 2.1 GPa), and n denotes the porosity of the specimen.

The results of the isotropic compression tests are summarized in Table 1. It can be noted that the rr-Hertz model with a realistic shear particle stiffness (i.e. $G_g = 29$ GPa) produces an unrealistically high soil skeleton bulk modulus K_{sk} , resulting in a B -value apparently below 1.0. Note that the B -value of a fully saturated granular soil should be very close to 1.0. In contrast, when the rr-Hertz model uses an artificially low stiffness (i.e. $G_g = 1.46$ GPa), the resulting K_{sk} and B -values are

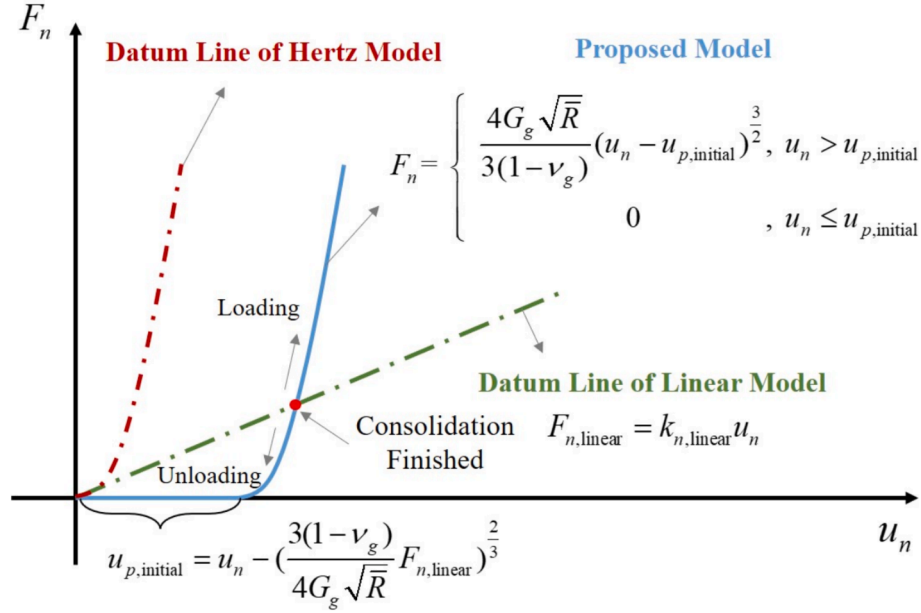


Fig. 1. The normal force–displacement law before abrasion accumulation.

comparable to both those based on the rr-linear model and experimental measurements of natural sands (Mitchell and Soga, 2005; Terzaghi and Peck, 1967).

This discrepancy illustrates a primary issue. While grain-scale experiments confirm that particle contacts are better described by the Hertz model and its high corresponding stiffness (Cole and Hopkins, 2017; Yao et al., 2022), these models fail to capture the bulk compressibility of the soil skeleton. Although using the rr-Hertz model with G_g of 1.46 GPa results in a soil skeleton bulk modulus comparable to the measured value of natural sands, it cannot reproduce the grain scale constitutive behavior. This inconsistency leads to the postulation that surface degradation, specifically the cumulative influence of particle abrasion, represent the significant factor that has been largely overlooked in conventional undrained DEM simulations. While factors such as particle geometry and fragmentation are undoubtedly primary drivers of soil response, physical and numerical evidence suggests that abrasion is notable during shearing and may play a crucial role in suppressing dilation (Sandeep and Senetakis, 2018; De Bono et al., 2020). This dilation reduction is, in turn, critical for capturing correct pore pressures during undrained simulations.

By potentially underestimating the cumulative deformation introduced by surface degradation, current simulations might not yet fully provide a physically sound representation of the soil skeleton's bulk compressibility. This is supported by single-particle and one-dimensional compression tests on quartz sand, which does not significantly fragment at low confining pressure, identifying particle rearrangement associated with abrasion as one of the important causes of soil stiffness variation in real tests (Cavarretta et al., 2010; Altuhafi and Coop, 2011; Cavarretta and O'Sullivan, 2012). Conventional undrained DEM simulation consistently neglects particle surface degradation, thereby underestimating the cumulative deformation in physical experiments. The linear contact model provides reasonable predictions only because its artificially low stiffness by calibration implicitly accounts for all stiffness-reducing mechanisms, including abrasion. The central challenge addressed in this study is to achieve macroscopic realistic simulations by explicitly accounting for the evolution of soil skeleton compressibility driven by surface degradation mechanisms (specifically abrasion).

3. A unified contact model with abrasion effect

3.1. Normal force–displacement law of the proposed model

To capture the mechanical behavior of granular soils across varying strain levels, a unified contact model is proposed that establishes particle abrasion as the predominant mechanism of surface degradation. Specifically, the model explicitly integrates the effects of abrasion into the normal force–displacement law to account for the enhanced compressibility observed in the soil skeleton.

During the initial sample preparation and consolidation phases, the simple and computationally efficient rr-linear model is employed to account for the realistic soil skeleton bulk modulus. The normal contact force between particles is determined by the rr-linear model as

$$F_{n,linear} = k_{n,linear} u_n \quad (3)$$

where $F_{n,linear}$ is the normal contact force, u_n is the overlap between particles, and $k_{n,linear}$ is the linear reference normal stiffness. During the shearing stage, Eq. (3) serves as a reference state indicating the initial-ization of abrasion and a limit for the post-abrasion normal force formulation.

The normal contact force law is designed to transition naturally and smoothly from a Hertz-like response at small strains to a more compliant, linear-bounded response at large strains, governed by the accumulation of abrasion. After consolidation, the contact model is switched to the proposed model, as illustrated in Fig. 1.

This transition establishes an initial abrasion displacement $u_{p,initial}$ for each contact, determined by

$$u_{p,initial} = u_n - \left(\frac{3(1-\nu_g)}{4G_g \sqrt{R}} F_{n,linear} \right)^{\frac{2}{3}} \quad (4)$$

where the contact radius $\bar{R}$ is defined as $1/\bar{R} = 1/R_1 + 1/R_2$. This initial offset accounts for the plastic deformation observed in grain-scale experiments when contacts are first loaded (Sandeep et al., 2019; Sandeep and Senetakis, 2018, 2019). This approach ensures a smooth transition with no abrupt change in contact forces.

Before the conditions for abrasion accumulation are met, the normal

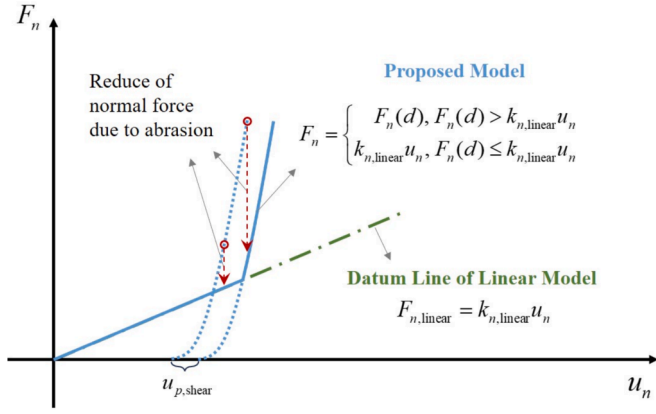


Fig. 2. The normal force–displacement law after abrasion accumulation.

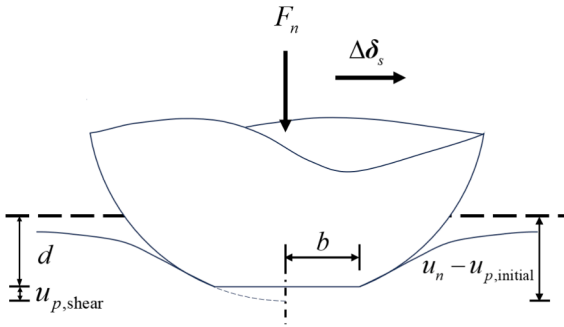


Fig. 3. The schematic of the contact between a flattened parabolic profile and flat plate.

force follows a Hertzian formulation that incorporates the initial abrasion displacement, indicated by the solid line in Fig. 1 as

$$F_n = \begin{cases} \frac{4G_g\sqrt{R}}{3(1-\nu_g)}(u_n - u_{p,initial})^{\frac{3}{2}}, & u_n > u_{p,initial} \\ 0, & u_n \leq u_{p,initial} \end{cases} \quad (5)$$

the corresponding normal stiffness is

$$k_n = \frac{dF_n}{du_n} = \begin{cases} \frac{2G_g\sqrt{R}}{1-\nu_g}(u_n - u_{p,initial})^{\frac{1}{2}}, & u_n > u_{p,initial} \\ 0, & u_n \leq u_{p,initial} \end{cases} \quad (6)$$

During the specimen shearing process, the particle normal force displacement law conforms to Eq. (5) until the following conditions for abrasion accumulation are met. Abrasion accumulation is initiated at a contact only when two conditions are met simultaneously: (1) the current normal force exceeds the linear reference force (i.e. $F_n > k_{n,linear} u_n$) and (2) the contact has previously experienced slip. Two typical contacts that satisfy the abrasion start conditions are marked with circles in Fig. 2.

Once initiated, the accumulation of volume loss is calculated at each timestep based on the Archard abrasive framework (Archard, 1953),

Table 2

Input model parameters in the simulation.

Model property	Value	Source
Linear reference normal stiffness $k_{n,linear}$	$k_0 \times r$	Gu et al., 2020b
Linear reference shear stiffness $k_{s,linear}$	$\alpha \times k_{n,linear}$	Gu et al., 2020b
Particle shear modulus G_g	29 GPa	Ng and Petrakis, 1996
Particle Poisson's ratio ν_g	0.15	Ng and Petrakis, 1996
Friction coefficient μ	0.45	Gu et al., 2020b
Rolling friction coefficient μ_r	0.5	Gu et al., 2020b
Abrasion coefficient K	10^4	Trial and error

Note: $k_0 = 8 \times 10^8 \text{ N/m}^2$, $\alpha = 0.15$ and r is the radius of the particle.

Table 3

Small-to-medium strain simulation program of CD tests.

Void ratio / Relative density	Confining pressure (kPa)
0.734 / 37%	50
0.696 / 50%	50
0.658 / 64%	50
0.731 / 38%	100
0.694 / 51%	100
0.657 / 64%	100
0.726 / 40%	200
0.691 / 52%	200
0.654 / 65%	200
0.717 / 43%	400
0.682 / 55%	400
0.647 / 67%	400

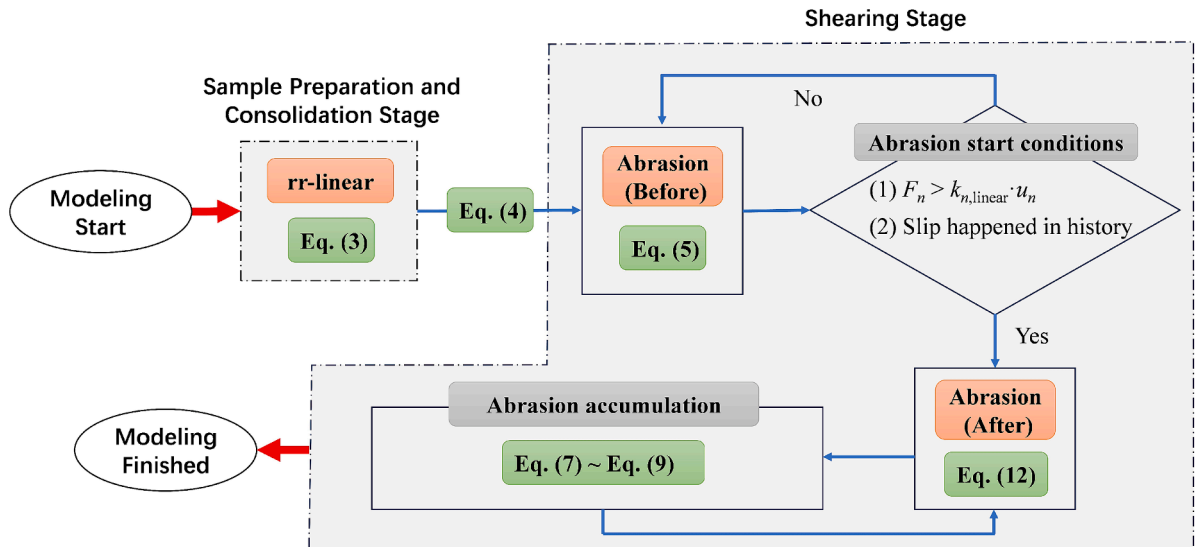


Fig. 4. Flowchart of the normal force–displacement law calculation.

Table 4
Large strain simulation program.

Void ratio / Relative density	Confining pressure (kPa)	Test type
0.726 / 40%	100	CD
0.603 / 83%	100	CD
0.728 / 40%	400	CD
0.609 / 81%	400	CD
0.731 / 38%	100	CU
0.661 / 63%	100	CU
0.733 / 38%	1000	CU
0.658 / 64%	1000	CU

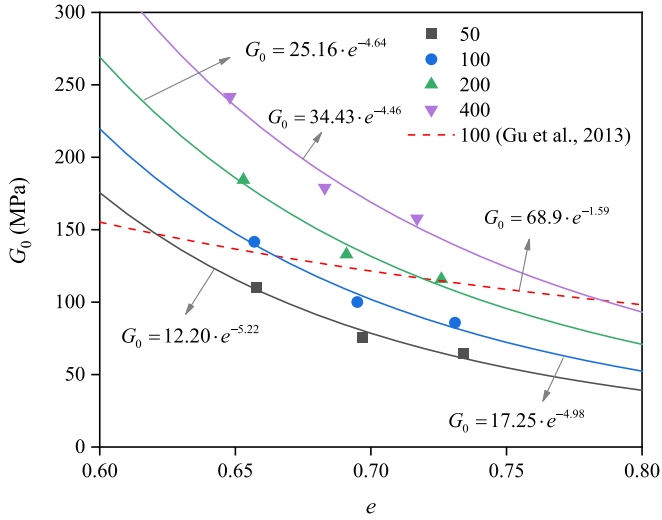


Fig. 5. Evolution of G_0 with e under different confining pressures.

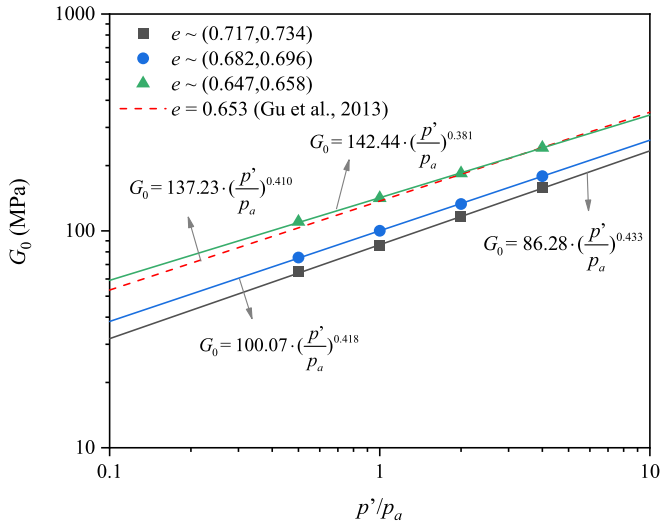


Fig. 6. Variations of G_0 with p'/p_a under different void ratio.

which is proportional to the tangential friction length and normal load, and inversely proportional to the surface hardness. Although the Archard framework is originated in metallurgy, applying it to granular soils is an active and accepted line of inquiry in recent DEM studies (De Bono et al., 2020; Capozza and Hanley, 2021). Specifically, the original form is used in this paper as

$$\Delta V = K \frac{F_n \|\Delta \delta_s\|}{G_g} \quad (7)$$

where ΔV is the damaged volume by abrasion, K is the abrasion coefficient, and $\Delta \delta_s$ is the relative tangential displacement. The validation of this form based on the Archard framework is presented in the Appendix A. This abrasion volume is then used to calculate an incremental reduction in normal displacement, which is accumulated as the shear-induced abrasion displacement $u_{p, \text{shear}}$, as illustrated in Fig. 2.

Normal displacement reduction $\Delta u_{p, \text{shear}}$ is determined using the geometric relationship as

$$\Delta u_{p, \text{shear}} = \frac{\Delta V}{\pi [\bar{R}^2 - (\bar{R} - u_{p, \text{shear}})^2]} \quad (8)$$

and the abrasion displacement is accumulated as

$$u_{p, t + \Delta t} = u_{p, t} + \Delta u_{p, \text{shear}} \quad (9)$$

where $u_{p, t}$ and $u_{p, t + \Delta t}$ are the total abrasion displacement at the current and next timesteps, respectively. As shown in Fig. 2, when abrasion accumulates, the nonlinear segment of the curve shifts to the right, accounting for the additional normal displacement introduced by the abrasion. The arrow indicates the decrease in normal force caused by abrasion.

After abrasion accumulates, the analytical solution for the normal contact force between a flattened parabolic profile and a flat plate can be found by the method of dimensionality reduction (Popov, 2010; Popov and Hess, 2016) as

$$F_n(d) = \frac{\sqrt{2} G_g b^3}{3(1 - \nu_g) \bar{R}} \left(2 + \sqrt{1 + \left(\frac{2\bar{R}}{b^2} d \right)^2} \right) \sqrt{-1 + \sqrt{1 + \left(\frac{2\bar{R}}{b^2} d \right)^2}} \quad (10)$$

where $b = \sqrt{\bar{R}^2 - (\bar{R} - u_{p, t + \Delta t})^2} \approx \sqrt{2\bar{R}u_{p, t + \Delta t}}$ is the half-width of contact plane and d is penetration depth defined as $d = u_n - u_p = u_n - u_{p, \text{initial}} - u_{p, \text{shear}}$. The corresponding schematic is shown in Fig. 3. The initial abrasion displacement $u_{p, \text{initial}}$ is obtained through Eq. (4) and $u_{p, \text{shear}}$ is the abrasion displacement during shearing stage.

The corresponding normal stiffness can be obtained by differentiating the normal force with the normal penetration as

$$k_n(d) = \frac{2\sqrt{2} G_g \bar{R} d}{(1 - \nu_g) b \sqrt{\sqrt{\frac{4\bar{R}^2 d^2 + b^4}{b^4}} - 1}} \quad (11)$$

Considering that the normal force is no less than the force obtained by the datum line of the linear model for reasonable macroscopic performance, the normal contact force and contact stiffness are determined by

$$F_n = \begin{cases} F_n(d), & F_n(d) > k_{n, \text{linear}} u_n \\ k_{n, \text{linear}} u_n, & F_n(d) \leq k_{n, \text{linear}} u_n \end{cases} \quad (12)$$

$$k_n = \begin{cases} k_n(d), & F_n(d) > k_{n, \text{linear}} u_n \\ k_{n, \text{linear}}, & F_n(d) \leq k_{n, \text{linear}} u_n \end{cases} \quad (13)$$

The normal force after abrasion accumulation governed by Eq. (12) is plotted as the solid line in Fig. 2, which is a combination of post-abrasion formulation and datum line of the linear model. Fig. 4 shows the flowchart of the determination of normal contact force during the entire simulation process.

3.2. Tangential force–displacement law of the proposed model

The tangential stiffness of a smooth sphere without slip is still used in the proposed model (Cundall, 1988; Nie et al., 2025; Otsubo et al., 2017). The tangential stiffness and normal stiffness at any time step satisfies the following relation

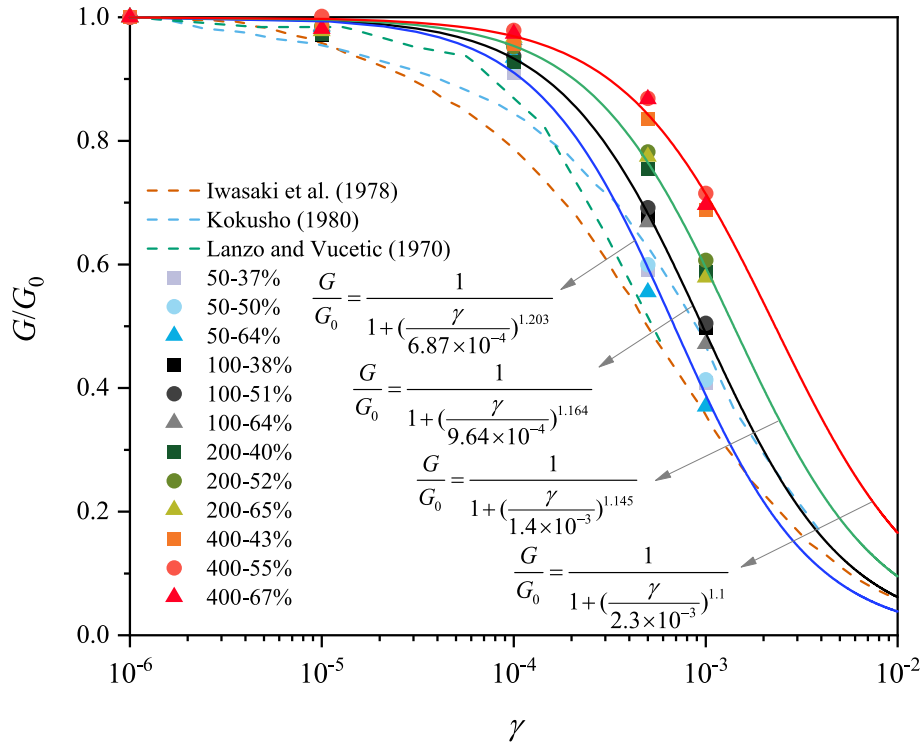


Fig. 7. The degradation curve of shear modulus G/G_0 with shear strain γ .

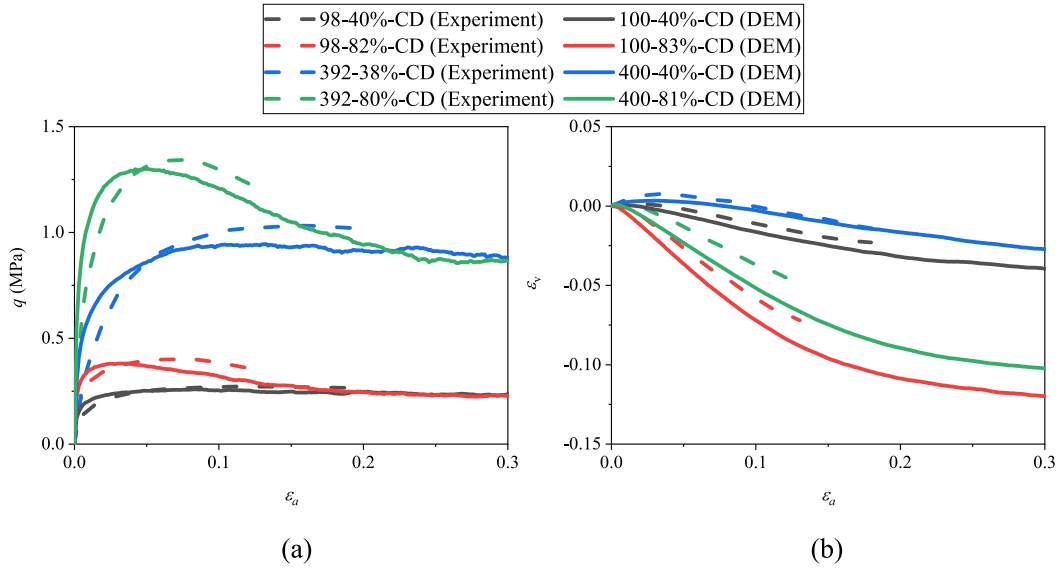


Fig. 8. Evolutions of (a) deviatoric stress q and (b) volumetric strain ϵ_v against axial strain ϵ_a in drained triaxial tests.

$$k_s = \frac{2(1 - \nu_g)}{2 - \nu_g} k_n \quad (14)$$

The tangential force between particles is updated by the following equation

$$\mathbf{F}_{s,t+\Delta t} = \mathbf{F}_{s,t} + k_s \Delta \delta_s \quad (15)$$

where $\mathbf{F}_{s,t}$ and $\mathbf{F}_{s,t+\Delta t}$ are the tangential contact forces at the current and next timesteps, and $\Delta \delta_s$ is the relative tangential displacement. The tangential contact force $\mathbf{F}_s$ is updated incrementally based on the relative tangential displacement $\Delta \delta_s$ and is limited by the Coulomb friction criterion $\|\mathbf{F}_s\| \leq \mu F_n$. The slip state of each contact is monitored, as slip is

a prerequisite for abrasion initiation.

3.3. Moment-angular displacement law of the proposed model

In the proposed model, the rolling resistance stiffness k_r is related to the linear referential tangential stiffness $k_{s,linear}$. This treatment allows a realistic modulus degradation at medium strains, while ensures the shear strength to be reasonable at large strains. The rolling resistance stiffness is determined as

$$k_r = k_{s,linear} \bar{R}^2 \quad (16)$$

The interparticle bending resistance moment is updated by the

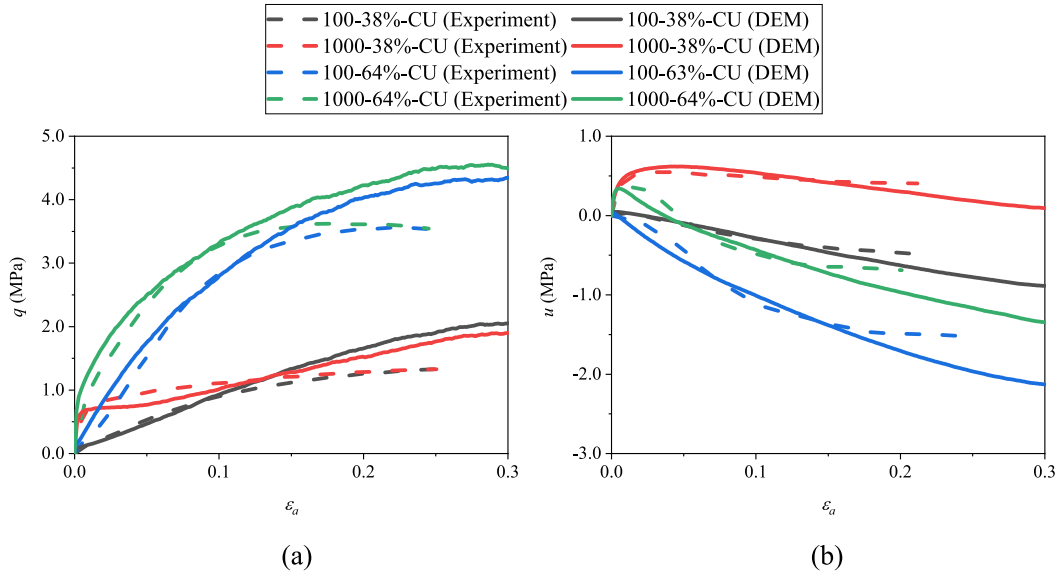


Fig. 9. Evolutions of (a) deviatoric stress q and (b) excess water pressure u against axial strain ε_a in undrained triaxial tests.

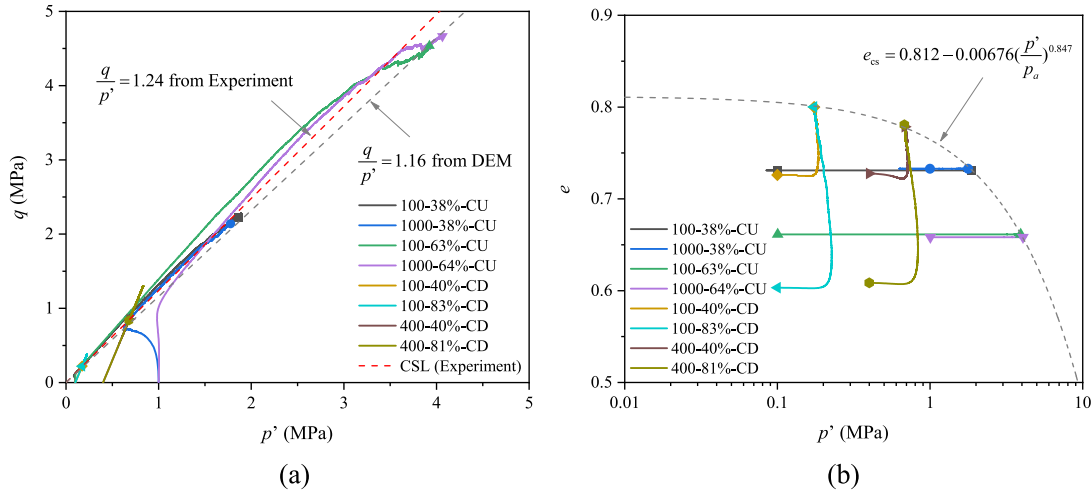


Fig. 10. Stress paths in (a) q - p' space and (b) e - p' space.

following equation

$$\mathbf{M}_{r,t+\Delta t} = \mathbf{M}_{r,t} + k_r \Delta \theta_r \quad (17)$$

where $\mathbf{M}_{r,t}$ and $\mathbf{M}_{r,t+\Delta t}$ are the bending moments at the current and next timesteps, and $\Delta \theta_r$ is the relative rotation angle. The interparticle bending moment $\mathbf{M}_r$ is updated incrementally based on the relative rotation angle $\Delta \theta_r$ and is limited by a Coulomb-type criterion $\|\mathbf{M}_r\| \leq \mu_r \bar{R} F_n$. Twisting resistance (Jiang et al., 2015) is not considered in this model. It is worth noting that the unified model remains stable throughout the entire simulation. The normal force remains unchanged after the switch. The tangential and rotational forces use an incremental approach, and their magnitudes and directions are preserved during the conversion.

4. Simulation results and discussions

4.1. Simulation program

During the simulation, the rr-linear model was used for sample preparation, pre-compression, and consolidation, following the specimen generation process detailed in Section 2. To achieve a range of

initial densities, the friction μ and rolling resistance μ_r coefficients were varied during the pre-compression stage, yielding a maximum void ratio $e_{\max}$ of 0.841 and a minimum void ratio $e_{\min}$ of 0.555 (Gu et al., 2020b; Wu et al., 2022). The consolidation was then performed with $\mu = 0.45$ and $\mu_r = 0.5$ and the void ratio after consolidation was calculated as the specimen's initial void ratio to accurately describe its pre-shearing state. For the subsequent shearing stage, the proposed unified model was activated. The complete set of input parameters for the proposed model is detailed in Table 2. Parameters from classical established models were retained consistent with those used in the literature (Gu et al., 2020b; Ng and Petrakis, 1996). The abrasion parameter K was obtained through trial and error against undrained experiments. The parametric study of abrasion coefficient K is presented in Appendix B. While the abrasion parameter K is rooted in the physical Archard abrasive framework, it functions as an effective parameter that encapsulates distinct surface degradation mechanisms, such as asperity breakage, smoothing, and wear, which are computationally prohibitive to model explicitly.

The simulation program included both drained (CD) and undrained (CU) triaxial tests. CD tests were conducted across small-to-medium and large strain ranges, while CU tests were simulated at large strains only using the constant volume method. Based on the Hertz model with a

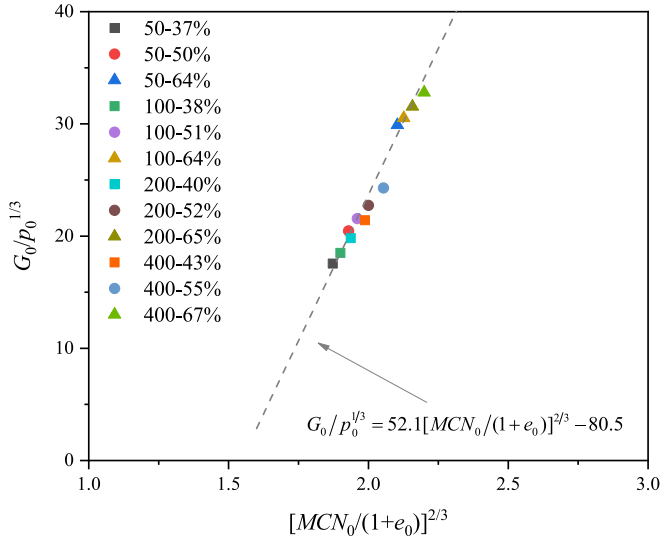


Fig. 11. Relationship between $G_0/p_0^{1/3}$ and $[MCN_0/(1+e_0)]^{2/3}$.

high particle shear modulus (i.e. $G_g = 29$ GPa), the simulation timestep is conservatively estimated as 2×10^{-8} s. To ensure quasi-static conditions, the inertia number is maintained less than 1.0×10^{-5} and 2.5×10^{-3} at small-to-medium and large strains, respectively (Da Cruz et al., 2005; Nguyen et al., 2017; Perez et al., 2016; Zhou and Xu, 2024). The specific conditions for each simulation are listed in Tables 3 and 4. All large strain simulations are terminated at an axial strain of 0.5 to ensure the critical state is reached.

4.2. Shear modulus at small strains

The small strain shear modulus G_0 is a fundamental parameter of soil and its dependence on void ratio and confining pressure can be analyzed using the empirical framework proposed by Hardin and Richart (1963)

$$G_0 = AF(e) \left(\frac{p}{p_a} \right)^m \quad (18)$$

where A is a fitting constant, $F(e)$ is a function related to the void ratio, p^* is mean effective stress, p_a is the reference stress (taken as 100 kPa in this study), and m is the stress component.

Fig. 5 shows the variation of the small strain shear modulus G_0 with initial void ratio e under different confining pressures. The void ratio function $F(e) = e^x$ is used to characterize the effect of the void ratio on small strain shear modulus (Lo Presti et al., 1997). The experimental data from Gu et al. (2013) at a confining pressure of 100 kPa is also included for comparison. It shows that the G_0 obtained at different void ratios is comparable to that obtained from physical tests. However, the exponential correlation derived from the simulations exhibits a steeper slope than experimental results. This is likely attributed to the use of spherical particles, which limits the achievable void ratio range compared to real, irregularly shaped soil particles.

The effect of confining pressure on G_0 is illustrated in Fig. 6. The simulation results show an excellent agreement with the experimental data (Gu et al., 2013), capturing the characteristic exponential increase in stiffness with rising confining pressure. The fitted power-law exponents from the simulations ranging from 0.381 to 0.433 align closely with the experimental value of 0.41, confirming the model's performance in representing stress-dependent modulus.

4.3. Nonlinear modulus degradation at intermediate strain

The degradation of shear modulus can be predicted by the empirical formula (Benz, 2007; Darendeli, 2001)

$$\frac{G}{G_0} = \frac{1}{1 + (\gamma/\gamma_r)^a} \quad (19)$$

where G is the corresponding secant shear modulus at a shear strain of γ , γ_r is the reference shear strain, and a is a dimensionless exponent.

Fig. 7 shows the degradation of the shear modulus under medium strains. The simulation results are compared with experimental data at a confining pressure of 100 kPa (Iwasaki et al., 1978; Kokusho, 1980; Lanzo and Vucetic, 1970). The simulated curve at 100 kPa is slightly higher than the experimental data but successfully captures the overall nonlinear trend. The result shows that the simulated modulus degradation is not highly sensitive to void ratio but is influenced by confining pressure, which is consistent with previous findings (Iwasaki et al., 1978; Kokusho, 1980; Senetakis et al., 2013). Specifically, the shear modulus degrades more slowly at higher confining pressure.

4.4. Critical state at large strain

The model's ability to capture large-strain behavior was rigorously evaluated by comparing simulation results from drained (CD) and undrained (CU) triaxial tests against classic experimental data. The plots of deviatoric stress and volumetric strain for drained tests in Fig. 8 show excellent agreement with the findings of Fukushima and Tatsuoka (1984). The model accurately captures key features, including the initial stiffness, peak shear strength, and post-peak strain-softening or strain-hardening behavior. Similarly, the evolution of excess pore water pressure in undrained tests (Fig. 9) closely mirrors the experimental data from Verdugo and Ishihara (1996), correctly simulating both positive pore pressure generation in contractive samples and negative pore pressure development in dilative ones. A key observation from the results is the model's adherence to expected soil mechanics principles. Specimens prepared at lower confining pressures or higher relative densities consistently exhibit more pronounced dilatancy under drained conditions and a stronger tendency toward developing negative excess pore pressures (strain hardening) under undrained conditions. The result demonstrates that the model correctly captures the influence of initial state (i.e. density and stress) on the subsequent shear response.

Fig. 10(a) shows that all stress paths, regardless of initial conditions or drainage, migrate towards a single, well-defined critical state line (CSL) in q - p' space. The slope of this line, representing the critical state stress ratio, is 1.16 in the simulations, which aligns well with the experimentally determined value of 1.24 (Verdugo and Ishihara, 1996). It is noteworthy that surface abrasion has a negligible effect on the critical state stress ratio for granular materials composed of quartz (De Bono et al., 2020; McDowell and Li, 2016). The state paths plotted in e - p' space (Fig. 10(b)) provide the compelling evidence of the model's performance for modeling the large strain behavior. The final states of all specimens clearly show a convergence towards a unique critical state line. This crucial validation confirms that the specimens' long-term behavior is independent of the initial state and is solely a function of the mean effective stress. The simulated CSL can be described by the following expression, consistent with established frameworks (Gu et al., 2014; Huang et al., 2014; Verdugo and Ishihara, 1996):

$$e_{cs} = 0.812 - 0.00676 \left(\frac{p}{p_a} \right)^{0.847} \quad (20)$$

The ability of the proposed model to reproduce these large-strain behaviors and converge to a unique critical state from various initial conditions provides strong evidence that the incorporated abrasion mechanism accurately captures the essential physics governing soil behavior at large deformations. The unified model incorporating abrasion effect proposed in this study employs a top-down calibration approach. Combining all results from Sections 4.2 to 4.4, the phenomenological parameters obtained through calibration yield favorable macroscopic outcomes and serve as an effective approach for simulating

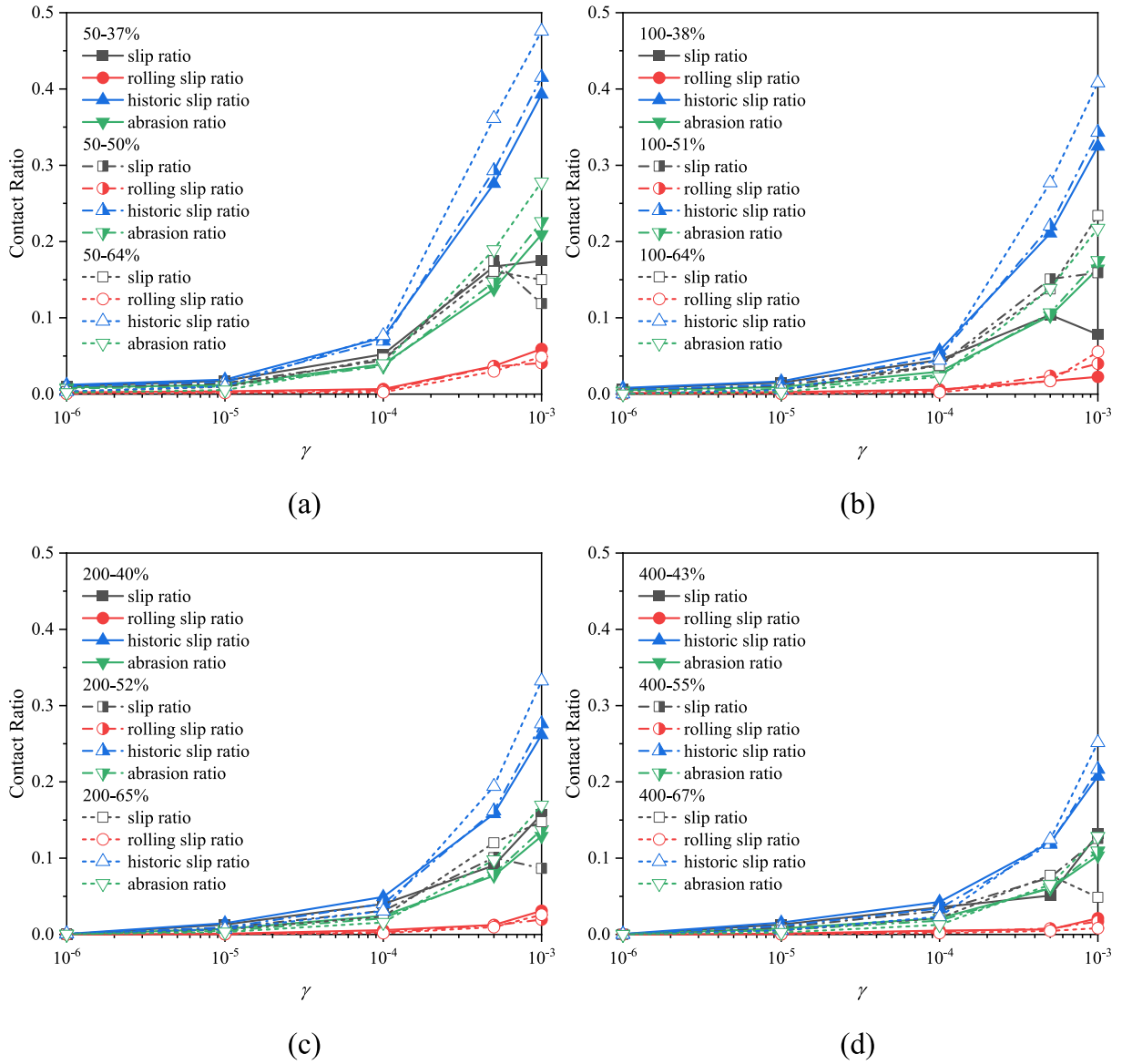


Fig. 12. Variations of different contact ratios at small-to-medium strain with confining pressure of (a) 50 kPa, (b) 100 kPa, (c) 200 kPa and (d) 400 kPa.

the mechanical properties of a granular soil across the full strain range.

4.5. Microscopic interpretation at small and intermediate strain

The macroscopic stiffness of a granular material is fundamentally governed by its internal fabric, particularly the connectivity of its particle contact network. This network can be quantified by the Mechanical Coordination Number (MCN), which excludes non-load-bearing particles (rattlers) and is defined by Thornton (2000) as

$$MCN = \frac{2C - N_1}{N - N_1 - N_0} \quad (21)$$

where C is the total number of particle contacts, N is the total number of particles in the sample, N_0 is the number of particles without contact, and N_1 is the number of particles with only one contact.

As indicated by micromechanical theory, the macroscopic stiffness should be directly related to MCN (Chang et al., 1989; Hu et al., 2023). Fig. 11 confirms this, showing a clear linear relationship between the stress-normalized shear modulus $G_0/(p_0)^{1/3}$ and the normalized initial coordination number $[MCN_0/(1 + e_0)]^{2/3}$. In this study, such linear relationship can be fitted by

$$G_0/p_0^{1/3} = 52.1[MCN_0/(1 + e_0)]^{2/3} - 80.5 \quad (22)$$

This establishes a direct link between the microstructure of the specimen and its small-strain elastic stiffness.

The transition from linear elastic to nonlinear behavior as strain increases is driven by changes at the contact level. Fig. 12 illustrates the evolution of different contact state fractions at small-to-medium strains. The slip ratio and rolling slip ratio represents the percentage of contacts that meets the tangential Coulomb friction criterion and rolling resistance criterion at the current state, respectively. The historic slip ratio tracks contacts that have ever slipped (met the tangential Coulomb friction criterion) in their history, which is a key precondition for activating abrasion in the proposed model. The abrasion ratio indicates the percentage of contacts where abrasion is actively accumulating. At very small strains (i.e. $\gamma = 1 \times 10^{-6}$), all ratios are negligible, indicating a predominantly elastic response. However, as the shear strain increases into the intermediate range (i.e. $\gamma > 1 \times 10^{-4}$), these ratios increase dramatically, signaling the onset of widespread particle sliding and abrasion. This microscopic transition is directly responsible for the macroscopic modulus degradation shown previously in Fig. 7. The results also show that higher confining pressure inhibits the growth of

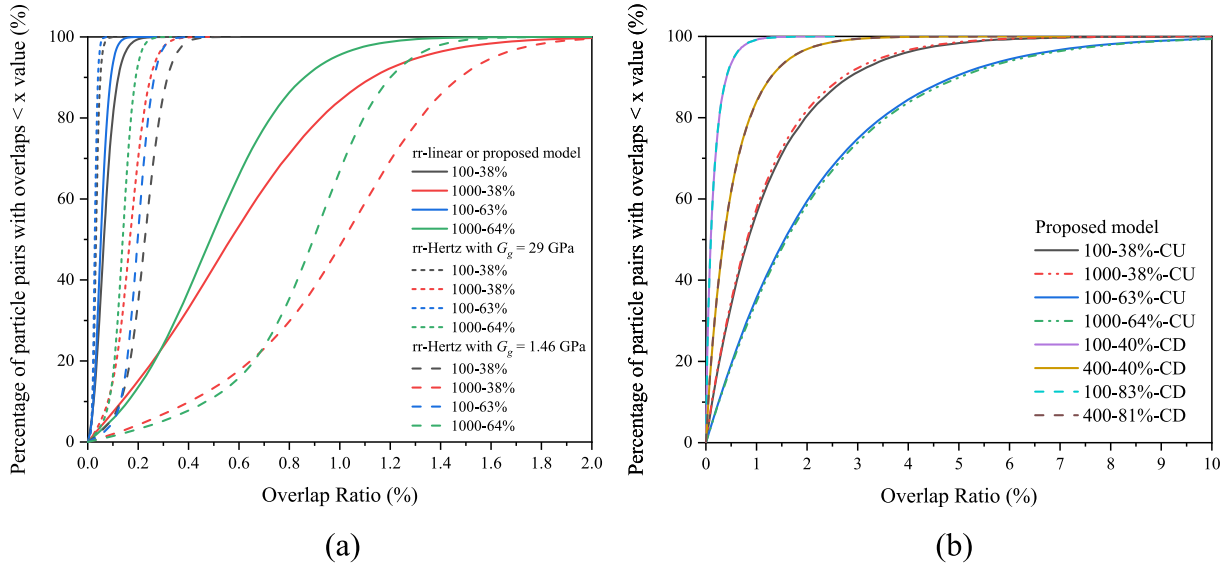


Fig. 13. The distribution of particle overlap after (a) consolidation and (b) after triaxial shearing.

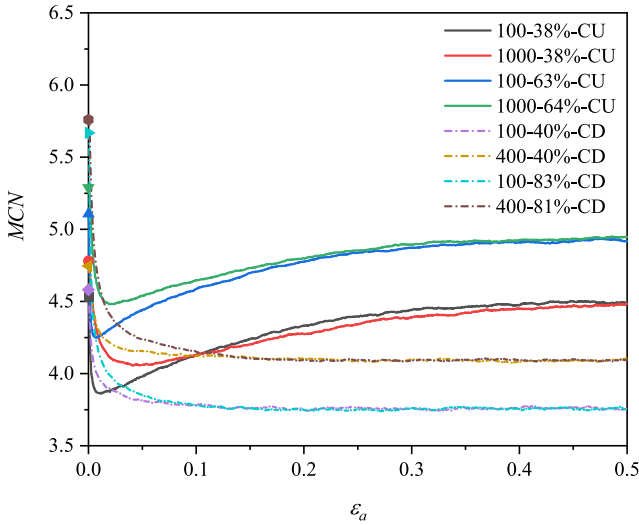


Fig. 14. The evolution of mechanical coordination number MCN against axial strain ε_a in drained and undrained tests.

historic slip and abrasion contact fractions, which explains why the soil degrades more slowly under higher pressure.

4.6. Microscopic interpretation at large strain

Fig. 13(a) shows the distribution of particle overlap after consolidation for specimens with different initial states. The results for rr-linear model and rr-Hertz model ($G_g = 1.46$ and 29 GPa) are also included for comparison. At the same confining pressure, specimens using the rr-Hertz model with a high particle modulus ($G_g = 29$ GPa) exhibit the smallest overlaps. This confirms that this particle system is stiffest, which is consistent with the findings in Section 2. Fig. 13(b) presents the distribution of particle overlap after the drained or undrained triaxial shearing. In both cases, over 90% of particle overlaps are less than 5% of the particle diameter. This small overlap satisfies the point-contact assumption essential for the discrete element method (Hanley et al., 2013).

Fig. 14 shows the evolution of MCN with axial strain ε_a in drained and undrained tests. As expected, the initial MCN (i.e. MCN_0) is higher

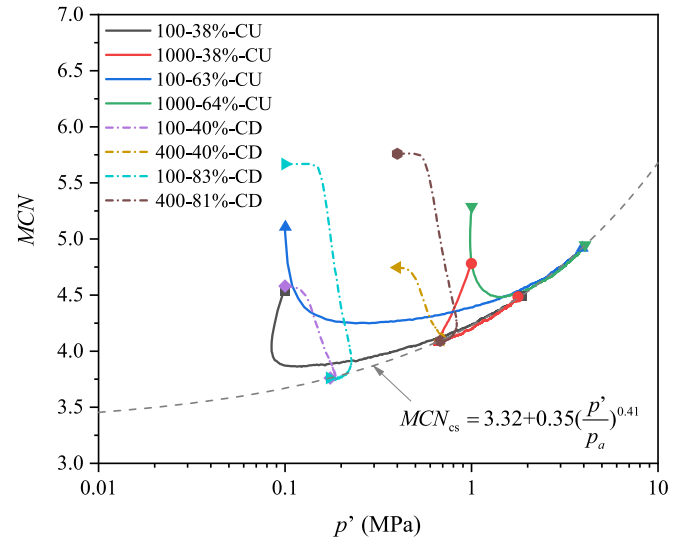


Fig. 15. Stress paths in MCN - p' space.

for specimens with greater relative density and confining pressure. During drained tests, the MCN decreases to a stable value. In contrast, during undrained tests, it first decreases and then increases to a critical

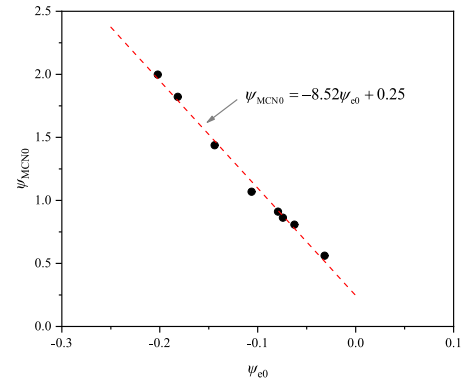


Fig. 16. Relationship between micro state parameter ψ_{MCN0} and macro state parameter ψ_{e0} .

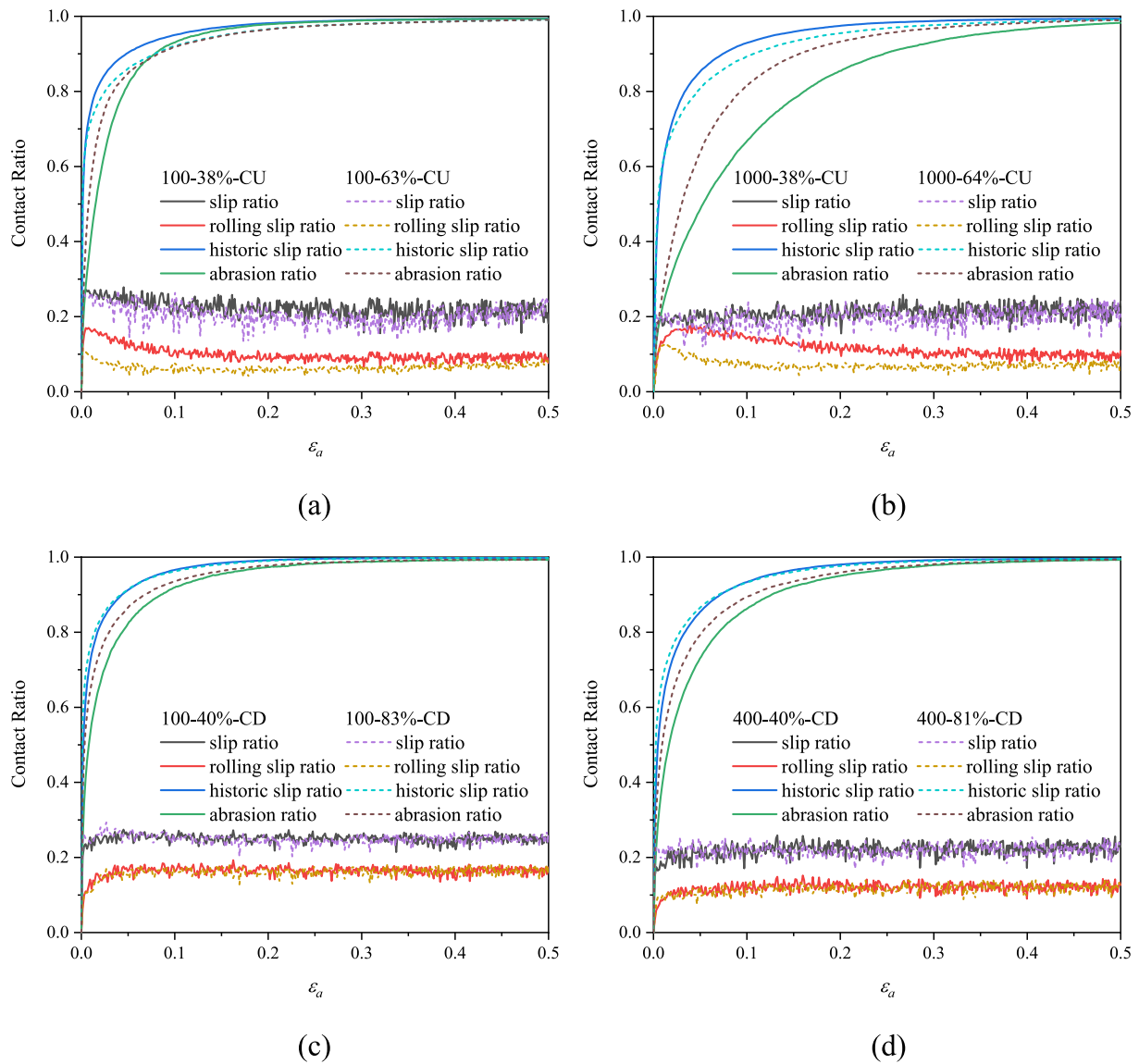


Fig. 17. Variations of different contact ratios at large strains with CU at (a) 100 kPa, (b) 1000 kPa and CD at (c) 100 kPa, (d) 400 kPa.

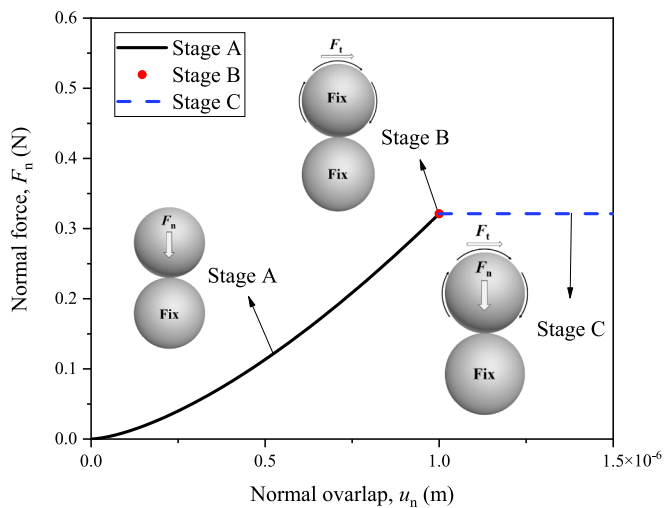


Fig. 18. Schematic for particle-scale numerical experiments.

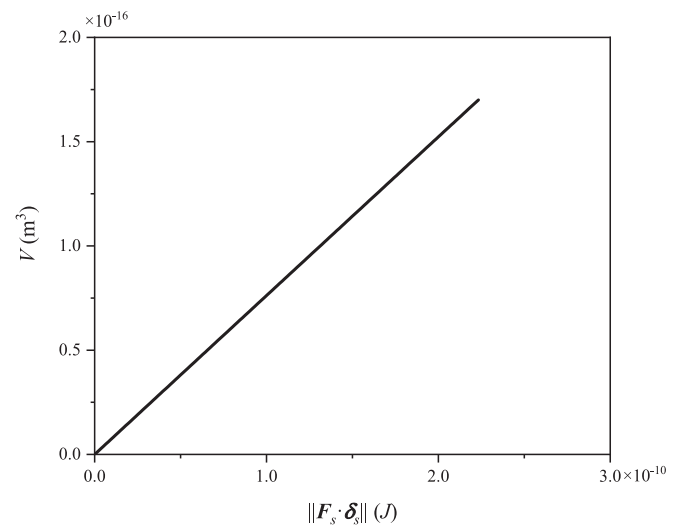


Fig. 19. The relationship between total loss volume V and frictional work $\|F_s \cdot \delta_s\|$ during Stage C.

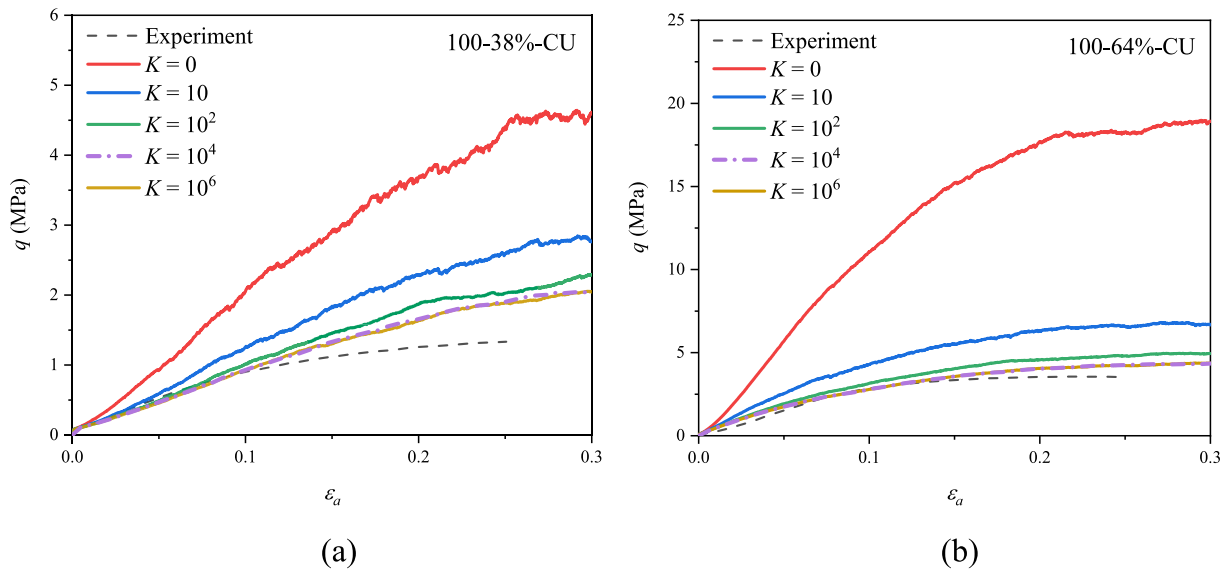


Fig. 20. Undrained simulation with varying abrasion coefficient K under 100 kPa with (a) $D_r = 38\%$ and (b) $D_r = 64\%$.

value, with the convergence conditions matching the observations in Fig. 13(b).

Fig. 15 maps the MCN against the mean effective stress p' . Previous work has shown that the critical state MCN (i.e. MCN_{cs}) forms a unique line in the MCN - p' space, independent of the initial sample state (Gu et al., 2020b; Huang et al., 2014). Here, this line is fitted as

$$MCN_{cs} = 3.32 + 0.35 \left(\frac{p'}{p_a} \right)^{0.41} \quad (23)$$

At large strain, both the critical state void ratio and the coordination number converge to unique lines, indicating a macro-micro correlation between them. The fitting relationship between initial micro state parameter $\psi_{MCN0} = MCN_{cs} - MCN_0$ and macro state parameter $\psi_{e0} = e_{cs} - e_0$ is shown in Fig. 16. Combining the results from Figs. 11 and 16, it can be concluded that both the small-strain modulus and critical state properties exhibit an inextricable relationship with MCN . It is worth noting that although other researchers have reported similar conclusions (Nie et al., 2022; Banerjee et al., 2024), these relationships are demonstrated using a single unified model that is valid across the entire strain range in this work.

Fig. 17 displays the fractions of contacts experiencing different states including slip, rolling slip, historic slip, and abrasion at large strains. The slip ratio and rolling slip ratio rise sharply at the beginning of loading, then evolve to fluctuate around a stable value. The slip ratio is larger than rolling slip ratio under both drained and undrained conditions. When the axial strain reaches 0.5, the historic slip ratio and the abrasion ratio for all specimens approach 1.0. This indicates that at the critical state, nearly all contacts are in a state of abrasion accumulation. It is also observed that an increase in confining pressure results in a slower accumulation of the abrasion ratio. By combining the results from Figs. 12 and 17, a clear pattern emerges. The contact abrasion ratio is relatively small at small-to-medium strains, especially when strain is less than 1×10^{-4} , but increases significantly at the large strain stage. This behavior implies that the proposed model performs similarly to a Hertz model at small strains while behaves more like a linear model at large strains. This dual characteristic allows the model to provide realistic results for discrete element analyses of sand across a wide range of strain magnitudes.

5. Conclusion

In this study, a unified contact model incorporating the effects of

particle abrasion is proposed to simulate the mechanical behavior of granular soils across small to large strain levels. The macroscopic behavior obtained from the simulations is compared with experimental data, for validating the proposed model and the adopted parameters. Further analysis of microscopic parameters reveals the mechanism of the macroscopic behaviors of granular soil during shearing from small to large strains. Both small strain modulus and critical state properties of the granular soil can be quantitatively linked to the mechanical coordination number. The main conclusions are as follows:

- (1) While the Hertz model is essential for capturing grain-scale stiffness, it faces challenges in reproducing soil behavior during monotonic undrained tests. This study postulates that the omission of surface degradation mechanisms that enhance the compressibility of the soil skeleton contributes to this discrepancy. Previous discrete element simulations using a linear model or nonlinear Hertz model had to adopt an unrealistically low contact stiffness or particle modulus to accommodate the compressibility of the soil skeleton at large strain.
- (2) A unified contact model explicitly considering abrasion effects is proposed, allowing for a natural transition from a Hertzian response at small strains to a linear-bounded behavior at large strains. The simulation results show that the proposed model is highly consistent with the experimental results from small to large strains (up to an axial strain of 0.2).
- (3) Microscopic analyses show that there is a linear relationship between normalized shear stiffness $G_0/(p_0)^{1/3}$ and coordination number $[MCN_0/(1 + e_0)]^{2/3}$ at small strains. At large strains, there exists a unique relationship between critical mechanical number MCN_{cs} and mean effective stress in the MCN - p' plane. Macro state parameter ψ_{e0} and micro state parameter ψ_{MCN0} maintain a unique relationship.

Further advancement of the proposed model necessitates research in several areas. A primary focus should be the incorporation of realistic particle shape to better characterize the granular skeleton of the particle aggregate. Concurrently, the representation of intergranular abrasion could be further refined by establishing criteria based on direct grain-scale experimental data, thereby more accurately reflecting its physical manifestation on particle surfaces.

CRediT authorship contribution statement

Jiuyang Zhou: Writing – original draft, Investigation. **Xiaoqiang Gu:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Conceptualization. **Hongwei Wu:** Writing – review & editing, Validation, Methodology, Conceptualization. **Jing Hu:** Writing – review & editing, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A.: Validation of the proposed Archard abrasive framework

To demonstrate the physical significance within our proposed model, we have supplemented a grain-scale numerical experiment (Fig. 18). This test involves two spheres under a constant normal force subjected to relative rotation to accumulate abrasion. After establishing contact (Stages A and B), a servo-mechanism maintains a constant normal force while applying a constant rotational velocity to induce steady abrasion (Stage C).

The relationship between total loss volume V and frictional work $||F_s \cdot \delta_s||$ during Stage C is shown in Fig. 19. This relationship aligns with the fundamental principle of the Archard framework, which is that the volumetric loss of particles due to abrasion is proportional to the friction work.

Appendix B.: Parametric study of abrasion coefficient K

A parametric study is conducted to explore the influence of abrasion coefficient K on undrained behaviors at large strains. The contact parameters and other conditions remain consistent with those in the paper, while only the value of K varies. Fig. 20 shows the influence of K on deviatoric stress of undrained shearing under 100 kPa with relative density of 38% and 64%. The results show that as K increases, the deviatoric stress progressively decreases. When the abrasion coefficient K exceeds 10^4 , further increasing has a negligible effect on the simulation results. This paper employs $K = 10^4$ for all simulations.

Data availability

Data will be made available on request.

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